

Progress report 6 of 10

LLM-Guided Neural Network Compilation for Real-time UAV Edge Inference

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The winner is the fastest plan that passed

Report 5 said which plans are legal.

This note says which legal plan is kept.

Several legal plans are compiled on the same machine: fixed presets, the heuristic, and the model's plan. Each run records whether compile succeeded, whether the outputs still match the untouched graph, and the latency percentile p50.

The chosen plan is the one with the lowest p50 among the plans that passed the numeric check. A plan that is faster but fails the check stays in the table. It is not chosen. If the model's plan has no rank, that is the reason, not a missing run.

p50 is the median latency of repeated runs on this machine. It moves when the QEMU load moves, so two runs from different sessions are not one experiment. The numeric check compares the rewritten network with the original FP32 graph.

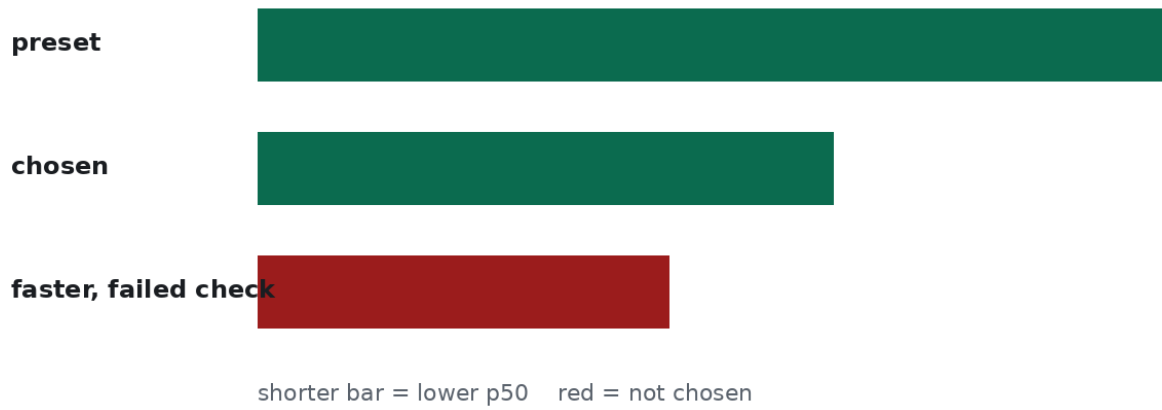


Figure 1. The short green bar can win. The red bar is faster and still rejected.

Speed without a passed check is not a win. The next note is where the chosen file is loaded.